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RESEARCH ARTICLE

IoMT-TrafficData: Dataset and Tools for Benchmarking Intrusion Detection in Internet of Medical Things

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ABSTRACT The healthcare industry relies heavily on a robust medical infrastructure but generates sensitive data about patients whose confidentiality and integrity protection must be guaranteed. But, although the Internet of Medical Things (IoMT) facilitates the interconnection of medical devices, software applications, and health systems, it also introduces vulnerabilities for adversaries to exploit. Moreover, in recent years, integrating machine learning (ML) into intrusion detection systems (IDS) have shown great potential in identifying malicious actions in the Internet of Things. However, such methods often require representative data for training, which is not commonly available for the IoMT. In this work, we introduce the *IoMT-TrafficData*, a dataset comprising IoMT network traffic data with features built over packet and network flow information for benign traffic and eight types of attacks. We present results from using several traditional ML algorithms and deep models to identify malicious traffic (binary classification) and the type of attack (multiclass classification), along with a comparative analysis of employing packet and flow statistics in ML-based intrusion detection. We show that ML algorithms can achieve high performance in identifying malicious traffic and distinct attacks, as most of the evaluated methods achieved an F1-score of over 90%. We also show that their performance on traffic-packets is, on average, almost 3% better for identifying malicious traffic than the individual attacks, and they achieve up to 5% better performance when dealing with traffic-flow statistics than when working on packed-based features. Hence, our experiments show the potential of using IoMT traffic flows in ML-based IDS and the usefulness of the *IoMT-TrafficData* dataset in such a context and present results that may be a benchmark reference for those who work with the dataset. The dataset can be openly accessed through the DOI 10.5281/zenodo.8116337.

INDEX TERMS Cybersecurity, deep neural network, Internet of Medical Things, intrusion detection system, machine learning.

I. INTRODUCTION

The term Internet of Things (IoT) describes a plethora of intelligent objects (e.g. sensors and actuators) [1], [2] connected through the Internet with the capability of exchanging information [3], resources [4], and most importantly, acting and reacting according to different situations when needed [2]. Given that the well-being of an individual is crucial for leading a prosperous and peaceful life [5], [6], it became imperative to explore the intersection of this domain with healthcare [7]. This integration, known as the Internet of Medical Things (IoMT), holds the promise of improving patient care [8], enhancing medical processes [7], and ushering in innovative possibilities for healthcare delivery [9]. However, the intrinsic sensitivity of medical data and the imminent threat of cyberattacks underscore the crucial need for robust security measures in IoMT systems [10], [11]. The security of IoMT systems

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assumes vital importance due to the potentially severe consequences of security breaches [12]. These breaches jeopardise patient privacy but also disrupt critical medical services and introduce risks to patient safety [10]. The multifaceted challenges and vulnerabilities inherent in IoMT further accentuate the significance of effective security mechanisms. The expansive attack surface, the diversity of device types, and the potential ramifications of successful attacks collectively create substantial risks [9], [12]. Conventional security approaches, including Intrusion Detection Systems (IDS) [13], have been employed to mitigate these threats. However, it is increasingly evident that traditional security measures exhibit limitations [14], thereby highlighting the need to explore innovative solutions such as the integration of machine learning (ML) into IDS [14], [15], [16]. Several works have suggested the utilisation of this technique in an extended context [17], [18], [19], [20], [21], [22].

The utilisation of ML-based IDS is notably prevalent in the context of IoT networks. Employing this technology within such environments is imperative for analysing extensive datasets across diverse settings [23]. It facilitates the correlation of events, identification of patterns, and detection of anomalous behaviour that might otherwise elude detection [24]. Deploying a ML-based IDS necessitates access to a high-quality dataset meeting specific criteria. This involves utilising a dataset representative of the IoT network in question [25], featuring data of good quality [26], and maintaining a balance between benign and malicious traffic [27]. Fulfilling these criteria is essential for achieving optimal results in implementing ML models and effectively distinguishing between benign and malicious traffic [19]. While datasets related to IoMT inherit the requirements of IoT datasets, existing datasets in this domain often fall short of meeting these essential criteria [28].

Moreover, the development of datasets within this domain is paramount. The healthcare industry has witnessed a significant rise in cyberattacks and ransom demands [29], making the need for new protective measures, including robust datasets for training ML-based IDS, more critical. Indeed, alarming statistics reveal the severity of the situation. For instance, 43% of healthcare organisations in the US falling victim to ransomware attacks. Among these attacked organisations, 76% experienced an average of three or more attacks, resulting in detrimental consequences for patient care, including loss of life. IoMT devices accounted for 21% of all ransomware attacks. Since IoMT has emerged as a prominent and rapidly growing technology field within the healthcare industry [6].

In this work, we present the *IoMT-TrafficData* [30] dataset, a new dataset with network traffic of an IoMT scenario. This dataset aims to facilitate forthcoming benchmarking research and aid in developing specialised IDS tailored for healthcare applications. The dataset includes benign traffic generated by protocols such as CoAP, MQTT, and HTTP and malicious traffic generated by eight distinct attacks, and comprises two different capture approaches, namely flows and packets. The packet-oriented approach contains data over individual segments that traverse the network, whereas the flows-oriented approach refers to data on sequences of network packets that share common characteristics or belong to the same communication session. Additionally, we present a comparative analysis of the performance achieved when using different ML methods, including traditional ML algorithms and deep neural networks (DNN), to identify and classify malicious operations over the two capturing approaches used (i.e., packets and flows-oriented).

As aforementioned, our motivation stems from the need for comprehensive datasets on IoMT network traffic data that researchers may use to develop and test intrusion detection effective systems for malicious traffic detection and attack identification. Indeed, there is a lack of sets for comparing packet-based and flow-based strategies. Therefore, this work makes the following key contributions:

- Presents a comprehensive and representative IoMT dataset, encompassing benign and malicious traffic from eight distinct attack scenarios, four protocols, and over ten device types.
- Performs a comprehensive comparative analysis of traditional ML algorithms and deep learning (DL) models, including an evaluation of packets-based and flows-based features, to elucidate the strengths and limitations of distinct solutions.
- Illustrates the efficacy of ML algorithms in detecting malicious traffic and diverse attack types, achieving high performance and highlighting their potential for accurate threat detection in IoMT environments.

The rest of the paper is organised as follows. Section II presents some background and related work. Section III outlines the dataset creation process. In Section IV, we present and discuss results on applying ML to identify anomalous traffic using several approaches. Finally, Section V concludes the paper and suggests future research directions.

II. BACKGROUND

The healthcare industry is being rapidly transitioning from paper-based systems to electronic health records (EHR) systems to offer efficient and cost-effective services [31]. Within EHR systems, the IoMT can integrate seamlessly, enhancing workflows within medical clinics and hospitals. Although IoMT offers remarkable benefits within the healthcare industry, it also inherits cybersecurity challenges from its parent concept of IoT [32], which are more critical due to the direct involvement of individuals. In these vulnerabilities, adversaries can exploit them to gain unauthorised access, compromising confidentiality, integrity, and potentially the availability of the IoMT system [28].

A. IOMT ATTACKS & INTRUSION DETECTION

Several authors (e.g., [9], [16], [19]) have noted a rising number of attacks to gain control over various devices within the IoT/IoMT ecosystem. They indicate that classifying the threats in the IoT/IoMT domain can be done based on the

targeted layer. Typical layers of IoT/IoMT include sensing, network, data management, and application layer [2]. For each layer, attacks such as Eavesdropping [33], Denial of Service (DoS) [19], [33], Man-in-the-middle (MiTM) [34], and Reconnaissance [9], [16] are examples of potential attacks that can occur at each respective layer.

A DoS is an attack that targets the availability of a system or network by overwhelming it with illegitimate requests. Variations of this attack include Apache Killer, R-U-Dead-Yet, Slow Read, and Slow Loris [35], [36]. In ARP Spoofing, attackers can potentially intercept sensitive medical data transmitted between devices by falsifying MAC addresses and redirecting network traffic [36], [37]. MQTT Malaria is similar to a DoS attack, which involves flooding the MQTT broker with illegitimate requests [38]. Within the IoMT domain, MQTT Malaria can disrupt the communication between medical devices, sensors, and back-end systems. In Reconnaissance, attackers gather information about the IoMT ecosystem to identify vulnerabilities, weak points, or potential entry points for further exploitation [16], [35]. Another threat to network security is CAM Table Overflow, a MAC flooding attack where an attacker inundates a network switch with an excessive number of counterfeit MAC addresses [39].

An IDS plays a crucial role in safeguarding networked systems, especially in the context of IoMT. These systems are essential for mitigating and detecting a wide range of known attacks [13]. Several authors explored the intersection between IDS, ML, and IoMT (e.g., [14], [40], [41], [42]). However, most use somewhat old datasets with certain flaws and consist of generic IoT data rather than IoMT-specific data to evaluate their ML applications. Still, they suggest that using an ML-based IDS may lead to a reliable and stable network for the IoMT ecosystem [14], [42], [43].

One common approach in ML-based IDS is anomaly detection, in which ML models learn the normal behaviour of individuals or objects through data analysis, creating a so-called normal profile [22]. Any behaviour differing from this profile is flagged as abnormal, allowing for the detection of potential intrusions or security breaches. These IDS take various forms of input, including logs [44], packets [45], [46], and flows [44], [47].

In this work, we focus on network packets and flows. Packets refer to individual data segments that traverse the network from the sender's origin to the intended destination [45]. Each packet contains specific information, including source and destination address, and metadata extracted via deep packet inspection (DPI) [46]. We can gain insights into the accessed resources and their corresponding locations by analysing the metadata. On the other hand, flows refer to sequences of network packets that share common characteristics or belong to the same communication session [47]. They represent the data movement between source and destination devices, encapsulating information such as source and destination IP addresses, ports, protocols, and timestamps [44]. One must use a tool to convert the capture file into flow data. Tools like Zeek Flowmeter, Yet Another Flowmeter, and nProbe may generate flows from a capture file [18], [25], [42]. Typically, they create multiple files containing statistical information about the utilised capture.

B. RELATED WORKS

In this section, we will first review existing datasets. Then, we will review some works that apply ML methods over network data to identify malicious operations.

1) IOMT NETWORK TRAFFIC DATASETS

Representative annotated datasets with benign and malign network traffic generated in an environment encompassing various medical devices are required to evaluate the use of an ML-based IDS for IoMT. Such datasets must contain data from a network incorporating diverse IoMT-based data and sensors that reflect the characteristics of IoMT systems and architectures. The main datasets on network traffic for intrusion detection in IoMT scenarios are:

IoT-Healthcare - Hussain et al. [51] developed a utility named IoT-Flock to generate an IoT Healthcare Security dataset. This dataset simulates an IoT-based Intensive Care Unit (ICU) with a capacity of two beds, each equipped with seven environment monitoring devices and nine patient monitoring sensors. While being flows-based, the dataset comprises 188,694 samples, with 108,568 ($\approx$ 57.53%) representing benign traffic and 80,126 ($\approx$ 42.46%) representing malicious traffic. It is crucial to note that the dataset lacks a diverse range of attacks, and all the represented attacks are binary. Consequently, it is only possible to perform binary classification, making it unsuitable for training and evaluating multiclass intrusion detection models.

ICE - The Integrated Clinical Environment (ICE) dataset, introduced by Fernandez et al. [49], was generated from five medical devices, each running different operating systems, and one clinical database distributed across the machines with Microsoft Windows. The dataset aims to detect four types of ransomware attacks: Locky, SamSam, WannaCry, and Petya. Utilising the Netflow format and recording 17 different types of flow records, the authors collected 520 flows, consisting of benign and malicious data without specification. Consequently, the dataset lacks a satisfactory amount of data flows, which can pose challenges when evaluating it within ML applications for intrusion detection.

Wustl-EHMS - Developed by Hady et al. [50], the Wuslt-EHMS dataset was created using a real-time Enhanced Healthcare Monitoring System (EHMS) testbed. This testbed collects network flow metrics and patients' biometrics, such as oxygen levels, blood pressure, and temperature. The attacks performed within this dataset include two MiTM attacks, namely spoofing and data injection. The dataset consists of 16,318 samples, with 14,272 ($\approx$ 87.50%) representing benign traffic and 2,046 ($\approx$ 12.50%) representing malicious traffic. The dataset is labelled only for performing binary classification. While the scenario represents a real-world IoMT scenario, the performed attacks may not be

TABLE 1. Comparison of published datasets in the IoMT domain: key characteristics and essential features.

Year	Dataset	Packets	Flows	Binary Applicable	Multiclass Applicable	Wi-Fi & Bluetooth/BLE	CoAP & MQTT	Balanced Traffic	Attacks Rep. ($N \geqslant 5$)
1994	Diabetes [48]	-	✓*	✓	✓	-	-	✓	-
2019	ICE [49]	-	✓	✓	✓	Only Wi-Fi	-	-	-
2020	Wustl-EHMS [50]	-	✓	✓	-	Only Wi-Fi	-	-	-
2021	IoT-Healthcare [51]	-	✓	✓	-	-	Both	✓	-
2021	ECU-IoHT [52]	-	✓	✓	✓	Both	-	-	✓
2022	BlueTack [53]	-	✓	✓	✓	Only BLE	-	✓	-
2024	CICIoMT2024 [54]	-	✓	✓	✓	Both	Only MQTT	-	✓
2024	**IoMT-TrafficData** [30]	✓	✓	✓	✓	Both	Both	✓	✓

* *In the context of this dataset, "flows" refer to statistical representations of features, rather than network flow.*

representative or extensive enough for future intrusion detection evaluations. Moreover, the dataset's low number of flows and extreme imbalance between benign and malicious traffic pose additional challenges.

ECU-IoHT - Ahmed et al. [52] proposed the ECU-IoHT dataset for device security evaluations in the healthcare domain. During the dataset development, the authors employed standard methodologies for penetration testing. The healthcare testbed used was the Libelium MySignals healthcare kit, which collected data from temperature, blood pressure, and heart rate sensors. The authors launched five different attacks, namely: Nmap, ARP spoofing, DoS, Smurf attack, script injection, and bettercap attack. The dataset comprises 111,207 flows, with 87,754 ($\approx$ 79.91%) representing malicious traffic and 23,453 ($\approx$ 21.09%) representing benign traffic. Moreover, the dataset includes the type of attack for each malicious traffic, facilitating multiclass classification. However, the dataset lacks representative IoT/IoMT protocols, containing only standard protocols. Additionally, not only the dataset is unbalanced, but it consists of only nine features: flow number, record time, source IP, destination IP, protocol, length of the message, info, traffic type (normal or attack), and type of attack. While these features are essential for packet analysis and flow classification, they may not be suitable for post-ML evaluation, as they contain biased values such as source IP, destination IP, and info, which specifically inform the IP address of the communication and the port used.

Diabetes - Developed in 1994 by Kahn [48], this dataset records diabetes data from patients sourced from two origins: an automatic electronic recording device and paper records. It comprises only four features: date, time, code, and value, with the code variable having 20 different codes. This dataset is not designed to detect external intrusions but rather aims to identify anomalies in the usage of insulin pumps based on patients' diabetes records. While widely used for anomaly detection in the IoMT realm, due to its automatic electronic recording device, this dataset lacks a sufficient number of features, not fully representing IoMT devices and protocols.

BlueTack - The BlueTack dataset, proposed by Zubair et al. [53], includes data related to both Bluetooth Low Energy (BLE) and Basic Rate/Enhanced Data Rate (BR/EDR) technology. This dataset, collected over 52 hours and totalling over 4.4 GB of data, encompasses both normal traffic and malicious traffic resulting from various attacks. These attacks include DDoS, Bluesmack, and DoS on the L2CAP layer of the Bluetooth protocol. While serving as a representative dataset within the IoMT Bluetooth realm, it is relevant to note that the dataset solely focuses on the usage of Bluetooth within an IoMT scenario, which may not fully capture the complexity of real-world IoMT scenarios. Additionally, the attacks conducted in the dataset represent only a few types of possible attacks in the realm of Bluetooth, potentially limiting its applicability in assessing Bluetooth security comprehensively.

CICIoMT2024 - Released in 2024, the CICIoMT2024 [54] dataset is a comprehensive collection of data in the IoMT domain. The dataset was designed to establish a realistic benchmark for evaluating the security of healthcare devices connected to the IoMT. It comprises 18 distinct cyberattacks targeting 40 IoMT devices and contributes to the healthcare industry by incorporating a diverse set of protocols, including Wi-Fi, MQTT, and Bluetooth. Although the dataset is extensive in terms of attacks and devices used to simulate the IoMT environment, it lacks one of the most widely used protocols in the IoT/IoMT realm, namely the CoAP protocol. Furthermore, the dataset does not provide a methodology for utilising packet-level data to evaluate its benchmarks on ML models, instead offering only a flow-based solution.

An examination of Table 1, which provides a comparative analysis of the essential features, functionalities, and methodologies across various datasets, reveals that our dataset distinctively meets a set of crucial criteria. Specifically, our dataset satisfies the following requirements: *i*) it represents a realistic scenario, *ii*) it incorporates communication protocols that reflect real-world usage scenarios, *iii*) it provides a sufficient amount of data for each class, thereby reducing bias and overfitting without the need for data augmentation, *iv*) it includes representative attacks targeting critical vectors in the scenario, *v*) it supports both packet-based and flow-based analysis, and *vi*) it comprises a sufficient number of features to facilitate a robust evaluation of ML-based IDS.

TABLE 2. Published studies employing ML for security within an IoMT framework. If dataset information is omitted, an independent dataset was created and is inaccessible. Highlighted datasets pertain to IoMT.

Year	Study	Data Type	Best Clf.	Class. Type	Accuracy	Precision	Recall	F1-Score	Dataset Used
2019	Alrashdi et al. [55]	Flows	OS-ELM	Binary	98.19%	-	-	-	NSL-KDD [56]
2019	Newaz et al. [57]	Packets	HGuard	Binary	90.90%	91.00%	91.00%	91.00%	-
2019	Fernandez et al. [49]	Flows	OC-SVM	Binary	-	92.32%	99.97%	95.96%	**ICE** [58]
			NB	Multiclass	99.60%	-	-	-	
2021	Hady et al. [50]	Flows	ANN	Binary	90.04%	-	-	-	**Wustl-EHMS** [59]
2021	Saheed et al. [60]	Packets	RF	Binary	99.76%	99.75%	96.45%	99.45%	NSL-KDD [56]
2021	Said et al. [61]	Flows	SVM	Multiclass	95.71%	-	-	96.24 %	-
2021	Ahmed et al. [52]	Flows	INFLO	Binary	-	-	-	< 95.00%	**Ecu-IoHT** [62]
2021	Hussain et al. [51]	Flows	RF	Binary	99.51%	99.70%	99.79%	99.65%	**IoT-Healthcare** [63]
2021	Li et al. [64]	Packets	ANN	Binary	93.27%	-	-	-	NSL-KDD [56]
2022	Manoharan et al. [65]	Flows	DAE	Binary	95.05%	97.20%	97.20%	97.20%	**Diabetes** [48]
				Multiclass	95.70%	95.00%	100%	96.00%	
2022	Zubair et al. [53]	Packets	MLP	Binary	99.80%	99.70%	99.06%	99.38%	**BlueTack** [66]
				Multiclass	99.55%	91.12%	81.89%	83.50%	
2022	Ghourabi [67]	Flows	LightGBM	Binary	100%	100%	100%	100%	Edge-IIoT [25]
				Multiclass	-	92.00%	88.00%	89.00%	

2) ML-BASED WORKS

Alrashdi et al. [55] introduced a fog-based attack detection framework that employs an ensemble of online sequential extreme learning machines (EOS-ELM) to efficiently detect malicious activities. The authors conducted a comparative evaluation of their framework with other approaches such as ELM, DT, SVM, NB, and OS-ELM. An evaluation was performed using the NSL-KDD dataset, where all methods achieved results higher than 85% accuracy. Notably, EOS-ELM outperformed other methods, achieving an accuracy of 98.19%.

The study by Newaz et al. [57] proposes an ML-based framework, HealthGuard, for smart healthcare systems. HealthGuard integrates four distinct ML methods, namely ANN, DT, RF, and KNN. To evaluate the proposed framework, the authors curated their dataset, comprising data collected from eight different smart medical devices across twelve benign events, including seven normal user activities and five disease-affected events. The results indicate that HealthGuard achieved an accuracy of 91%.

Fernandez-Maimo et al. [49] introduced an automatic, intelligent, and real-time system designed to detect, classify, and mitigate ransomware in integrated clinical environments. The authors utilised the ICE dataset [58] to evaluate their proposed solution. Multiple ML algorithms were employed for both binary and multiclass classification tasks. The experiments yielded a precision and recall of 92.32% and 99.97%, respectively, in anomaly detection, and an accuracy of 99.99% in ransomware classification, demonstrating promising detection and mitigation times.

In their work, Hady et al. [50] focus on detecting malicious activity within a self-developed Enhanced Healthcare Monitoring System (EHMS) testbed. The primary objective of their research involves the detection of potential MiTM attacks, particularly those involving spoofing and data injection, within an IoMT scenario. Employing ML methods such as RF, KNN, SVM, and ANN, and comparing them across various types of features, their findings suggest that combining flow metrics and biometric data can enhance accuracy by up to 90.04%.

Several researchers have focused their studies on enhancing the security of insulin pumps [65], [68], [69]. These devices facilitate the automatic administration of insulin based on the patient's blood glucose levels. However, if compromised, insulin pumps pose a significant risk, as intruders could trigger potentially lethal insulin administrations [14]. Addressing this vulnerability, studies [15], [65], [69] aim to mitigate security gaps between components like the Carelink and insulin pumps. These components are vulnerable to two primary types of attacks: bolus dose attacks, involving the delivery of large insulin amounts in a short period, and basal dose attacks, where small but life-threatening insulin doses are administered over an extended period, jeopardising patient safety. To tackle these challenges, the application of supervised ML algorithms has been proposed. Specifically, SVM with a regression method was employed to learn typical insulin dosage patterns for each patient at different times of the day [15]. Through rigorous testing, it was observed that non-linear SVM outperformed linear SVM, achieving a 98% success rate in detecting single overdose attacks and demonstrating high efficacy in chronic overdose attack detection.

In 2021, Saheed [60] utilised the NSL-KDD dataset [56] to present a study demonstrating the effectiveness of deep recurrent neural network (DRNN) and supervised ML models in developing an efficient IDS for the IoMT environment. The study focused on classifying and forecasting unexpected cyber threats. Results showned that supervised ML models outperformed DRNN approaches, achieving an accuracy of 99.76%.

Said et al. [61] proposed an Anomaly Detection System (ADS) for a smart hospital IoT system, aiming to detect

events related to patients' health and environmental conditions, as well as network intrusions. Using Contiki Cooja as their simulation tool, they conducted experiments involving three different types of attacks. In multiclass classification, the proposed system achieved an accuracy of 91.71% and an F1-score of 96.24%. SVM emerged as the best classifier among the chosen ones.

In their work, Ahmed et al. [52] proposed the ECU-IoHT [62] dataset as a representative dataset for the IoMT. This dataset aims to evaluate various attacks and accurately distinguish malicious traffic from benign traffic. The authors tested the dataset using several ML algorithms, including KNN, LOF, COF, and INFLO, among others. The conclusions drawn indicated that the INFLO algorithm achieved an F1-score of over 95% in binary classification. It is noteworthy that multiclass classification was not explored in this study.

Hussain et al. [51] introduced a framework aimed at developing IoT context-aware security solutions for detecting malicious traffic in healthcare IoT scenarios. This framework incorporates an open-source IoT data generator tool called IoT-Flock. Using IoT-Flock, the authors generated a comprehensive IoT healthcare dataset containing both normal and IoT attack traffic. Subsequently, various ML techniques were applied to this dataset for cyberattack detection. The research covered four types of cyberattacks: MQTT Publish Flood, MQTT Authentication Bypass Attack, MQTT Packet Crafting Attack, and CoAP Replay Attack. Employing ML algorithms such as RF, NB, KNN, AB, LogR, and DT, and after thorough performance testing, the authors concluded that RF emerged as the most effective algorithm, achieving an impressive accuracy of 99.51% and an F1-score of 99.65%.

In 2021, Li et al. [64] introduced a method for intrusion detection based on ANN, where they utilised the butterfly optimisation algorithm, a meta-heuristic optimisation approach, to select optimal features for the learning process within the ANN. The results obtained, with an accuracy of 93.27%, demonstrate the capability of the butterfly optimisation algorithm in identifying discriminative features from network traffic data. Notably, the proposed algorithm outperformed other algorithms such as DT, SVM, and ant colony optimisation.

Manoharan et al. [65] proposed the utilisation of an IDS based on unsupervised ML to safeguard insulin pump injections against four types of generated attacks: Long Resume, Single Acute Overdose, Underdose, and Chronic Overdose. To achieve this, the authors introduced a Deep Autoencoder (DAE) and a DL algorithm to classify the logs generated by the medical equipment. They tested their hypotheses using the publicly available Diabetes [48] dataset and evaluated the model's performance in terms of accuracy, precision, recall, and F1-score. The results obtained surpassed those of traditional ML methods and prior research in this domain, achieving an accuracy of 95.05% for binary classification and 95.70% for multiclass classification.

Zubair et al. [53] introduced a decentralised, predictive, DL-based approach to autonomously detect and block malicious traffic, providing end-to-end defence against network attacks in IoMT devices, with a particular focus on Bluetooth technology. Additionally, the authors presented the BlueTack dataset [66] specifically designed for Bluetooth-based attacks against IoMT networks. The results for binary classification achieved an accuracy of 99.80%, while for multiclass classification, the accuracy reached 99.55%.

Regarding the necessity for malware detection in data servers and medical staff computers, Ghourabi [67] proposes not only a malware detection system but also an intrusion detection system. Utilising the Edge-IIoT dataset [25] and their proposed system based on the optimised LightGBM model, the study achieved a remarkable F1-score of 100% in binary classification and 89% in multiclass classification.

Table 2 provides an overview of the aforementioned studies. Based on the provided table, it is evident that our work distinguishes itself from previous studies by adhering to a set of essential criteria. We develop a representative and balanced IoMT dataset containing both benign and malicious traffic. Furthermore, we conduct an experimental evaluation of the dataset using various ML algorithms. Additionally, we provide a comparative and empirical analysis between the packets and flows approaches in ML-based intrusion detection.

III. THE IOMT-TRAFFICDATA DATASET

In this section, we present the process of creating the IoMT-TrafficData dataset, which comprises several steps. Figure 1 outlines the complete workflow deployed in this work: *i*) scenario construction, *ii*) attack generation and execution, *iii*) data collection and generation of flows, *iv*) dataset construction for packets, flows and Bluetooth, *v*) pre-processing with various techniques, *vi*) application of diverse ML models and assessment of the outcomes, and *vii*) analysis of the obtained results with the drawing of conclusions. The first four parts are detailed in this section, while the remaining ones, related to the malicious traffic and attack identification, are detailed in Section IV.

A. SCENARIO COMPOSITION

The scenario developed for this research represents a use case for a sports clinic. Within this scenario patients can monitor their performance, overall health, and several other parameters. For this purpose, we propose a straightforward yet representative scenario. Figure 2 illustrates the developed scenario, which has three main parts: *i*) Adversary Area, *ii*) General IoT Area, and *iii*) Wireless Body Area Networks (WBAN).

An adversary device was implemented and connected to the network within the first area *i*) to perform the cyberattacks. A General IoT Area, i.e. an area equipped with usual IoT devices, was developed in the second area *ii*). The devices implemented in this area were a PIR motion sensor and a DHT11 sensor, which integrates the functionalities of a temperature, humidity, and heat index

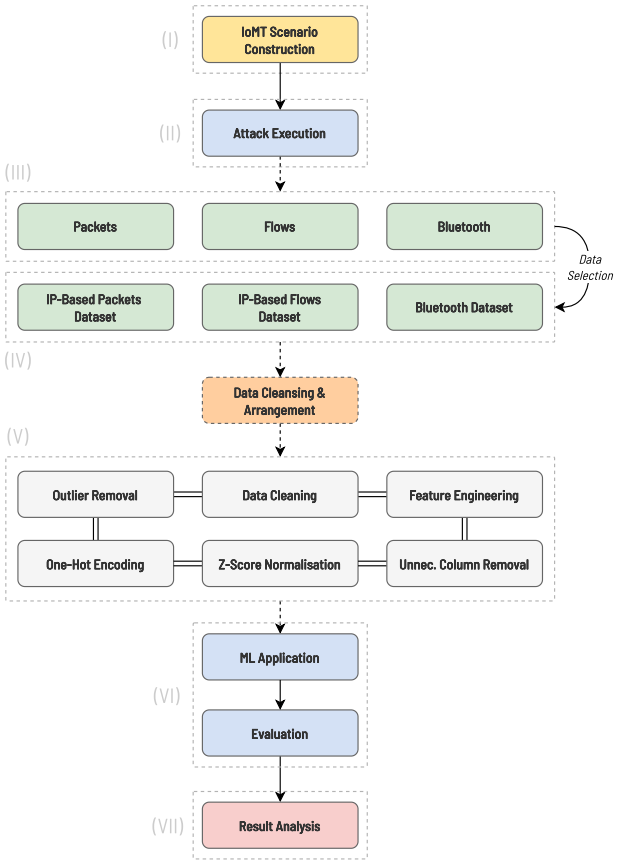


FIGURE 1. Deployed workflow for generating an IoMT representative dataset with ML application for intrusion detection.

sensor. We implemented two communication protocols to interconnect these sensors: MQTT and CoAP. An Arduino Yún, serving as an MQTT broker, receives communication from the PIR motion sensor. On the other hand, the Raspberry Pi, serving as a CoAP server, receives communication from the ESP32, and connected to the DHT11 sensor. The same Raspberry Pi also serves as an HTTP server, supported by the Apache technology. The communication occurs continuously, with a specific time interval between each information fetch. In the third area *iii*), devices related to WBAN are implemented. These devices comprise a smartwatch and a heart rate belt. The devices share data and establish communication with a smartphone using Bluetooth 5.0 and Bluetooth 4.0 Low Energy (BLE), respectively. It is worth noting that the smartwatch also communicates via Wi-Fi with the scenario gateway. Moreover, we also implemented a router and a switch. The router was necessary to aggregate the network into a private one to conduct all tests without external noise, and we implemented the switch to enhance performance and throughput.

B. ATTACK GENERATION

Various attacks were performed on the scenario deployed to generate and capture malicious data for further analysis. The attacks performed include four variations of DoS attacks, namely Apache Killer, R-U-Dead-Yet, Slow Read, and Slow Loris. We included variations of the same attack due to the prevalence of DoS attacks in the IoT domain [70]. While each attack possesses unique specifications, incorporating variations enables a nuanced understanding of their differences. Notably, these attacks specifically target the Raspberry Pi. The other attacks executed towards the IoMT scenario encompassed ARP Spoofing, CAM Table Overflow, MQTT Malaria, Network Scanning, Bluetooth Reconnaissance, and Bluetooth Injection. Table 3 presents all the attacks performed, the attack target and the tools used to execute each attack.

TABLE 3. Attacks carried out in the IoMT scenario, including details on the employed tools and the targets of the attacks.

Attack Designation	Attack Target	Adversary Tool
Apache Killer R-U-Dead-YET Slow Read Slow Loris	Raspberry Pi	slowhttptest [71]
CAM Table Overflow	Switch	macof [72]
ARP Spoofing	Raspberry Pi	bettercap [73]
MQTT Malaria	Servers (Pi & Yún)	mqtt-malaria [74]
Scanning/Sniffing	All Scenario	nmap [75]
BT Reconnaissance	IoMT Devices	bettercap [73]
BT Injection	Heart Rate Belt	nrf [76]

C. DATA COLLECTION

Two approaches were employed to collect the data generated from the IoMT scenario. The first approach involved collecting benign (i.e., normal) traffic over a specific duration, approximately 120 hours. It was possible to capture approximately 3 million packets of benign traffic. The second approach involved collecting malicious traffic, i.e., traffic generated through an attack towards the IoMT scenario. Contrary to benign traffic, malicious traffic does not have a specific duration for each attack because the amount of data generated per second varies from one attack to another. Indeed, 15,400,843 malicious packets were generated over approximately 5 hours by eight different attacks. To deal with the discrepancy between the amount of packets generated for benign and malicious traffic, we specified a ceiling on the number of packets for some attacks that produced a very high amount of traffic.

Data collection was performed using the tool `tcpdump`, with the `-G` flag set to `3600`. This configuration enabled file rotation, creating a new file every hour. This tool was integrated into an own-made script [77]. The primary purpose of this script was to streamline the process of data collection and facilitate the management of various services operating on the devices of the scenario, including the MQTT broker and CoAP server. Nevertheless, the tool was used to generate capture files in the `pcap` format.

Considering the data capture towards the Bluetooth protocol, the mentioned script was not utilised. For capturing the Bluetooth traffic generated within the scenario, it was used a Bluetooth dongle, specifically the nRF52840 from

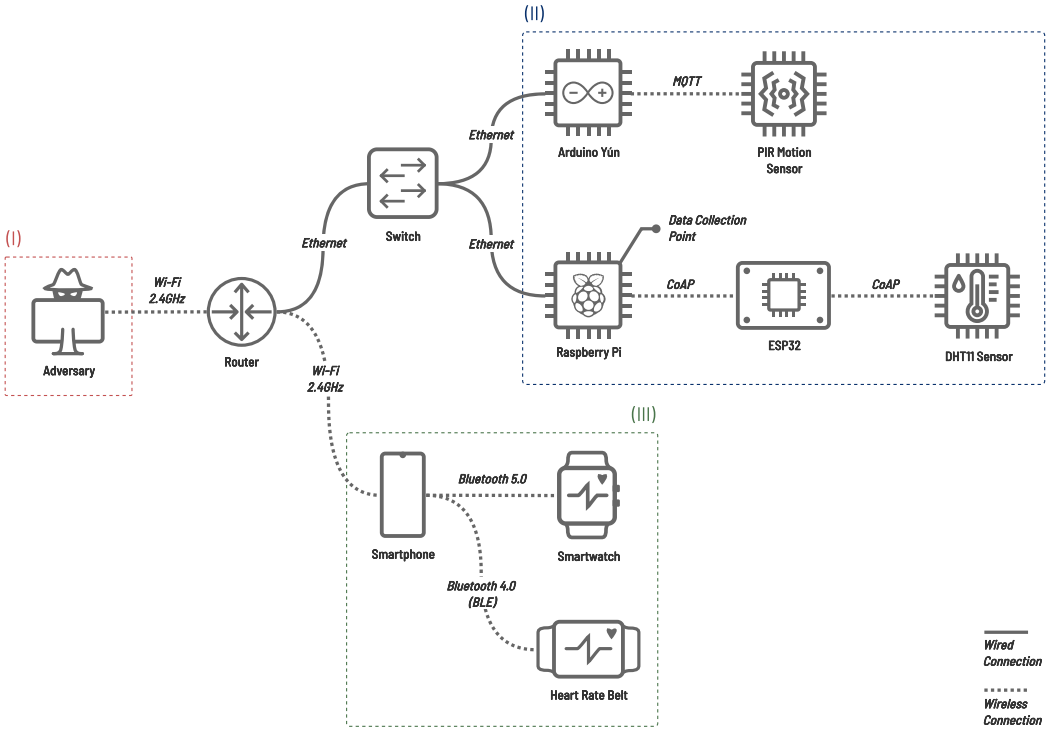


FIGURE 2. IoMT scenario developed within this research, focusing on the use case of a sports clinic. Solid lines represent wired connections, while dotted lines represent wireless connections.

Nordic Semiconductors. A total of approximately 974,630 packets were gathered, where 879,033 (90.17%) were from normal traffic, 43,812 (4.49%) were from the Reconnaissance attack, and 51,785 (5.21%) were from the Bluetooth Injection attack. In the forthcoming stages of the work, we used diverse techniques to balance benign and malicious traffic.

We place significant emphasis on the separation of the two distinct protocols, IP-based and Bluetooth. We assessed the similarity between the traffic generated by these protocols and observed that the traffic patterns from both protocols were entirely different. As a result, we decided to pursue two separate paths of work: generating a dataset specific to IP-based traffic and another dataset specific to Bluetooth traffic. This approach recognises the unique characteristics and requirements of each protocol, ensuring a focused and accurate representation of the respective traffic types.

Within the IP-based dataset, we also opted to follow two distinct approaches: one based on packets and the other based on flows, as we wanted to conduct a comparison between the results of these two approaches to understand which techniques produce better results. To generate the flows, we utilised the tool Zeek Flowmeter. As there were numerous capture files to convert to flows, we developed a script to automate this process. This script streamlines the generation of log files from capture files, providing a more efficient annotation and labelling workflow.

Within the packets approach, converting packets to CSV involves analysing the capture file and determining the necessary filters to extract all the information from it. To extract that information, we utilised the utility `tshark`. Due to the large number of fields to extract, we developed several scripts (available here) to assist us in this task. It is important to note that the development of the Bluetooth dataset follows the same workflow as the packets approach within the IP-based dataset.

D. DATA ARRANGEMENT & DATASET CONSTRUCTION

The *IoMT-TrafficData* consists of two sub-datasets: the IP-based dataset and the Bluetooth dataset. The IP-based dataset is further divided into two approaches: packets and flows. Due to the inherent differences among the three sub-datasets, their pre-processing techniques vary. However, the data cleaning process is consistent across both datasets.

This approach involved a detailed analysis of every feature within the dataset. The analysis was conducted using a methodology that experimented manually with the results derived from both DT and DNN models. For the DT, we used plot trees to understand the importance of each feature and determine whether it should be removed. For the DNN, we employed the Extra Tree Classifier, applying the same approach based on the importance weight of each feature.

1) IP AND FLOW-BASED DATA

For the IP-based dataset, within the flow approach, the development of the dataset involved seven different steps. These steps include data pre-processing, which involved removing redundant header rows from each log file, combining `conn.log` and `flowmeter.log` using the unique identifier `uid`, conversion to CSV format to facilitate data handling and analysis, data cleaning, focusing on

TABLE 4. Summary of capture duration and packet generation in IoMT scenario, along with record counts for packets and flows post pre-processing for all the sub-datasets: IP-based packets, IP-based flows, and Bluetooth.

Class	Capture Duration (Sec.)	No. Packets Generated (Approx.)	No. Records (Packets)	No. Records (Flows)
Normal (0)	576,00	3,032,261	3,032,261	300,000
Apache Killer (1)	1,800	1,077,195	376,990	2,334
R-U-Dead-YET (2)	1,800	780,895	375,942	71,990
Slow Read (3)	1,800	379,890	375,000	6,740
Slow Loris (4)	1,800	377,750	375,055	63,496
ARP Spoofing (5)	7,200	765,500	375,055	11,222
CAM Table Overflow (6)	600	2,398,320	379,419	62,500
MQTT Malaria (7)	900	6,573,400	379,837	4,961
Scanning (8)	900	3,287,890	379,407	62,500
Total of Instances (IP-Based)	114,000	15,382,341	6,048,938	3,243,185
Normal (0)	72,000	879,033	208,000	-
Bluetooth Reconnaissance (1)	108,00	43,812	104,000	-
Bluetooth Injection (2)	108,00	51,785	88,000	-
Total of Instances (Bluetooth)	114,000	974,630	400,000	-

removing non-network-originated data, labelling traffic as either malicious or benign for supervised learning, integration of the dataset into a single CSV file, and cleaning and validation of the final dataset to ensure data integrity and quality.

2) IP-BASED PACKETS AND BLUETOOTH DATA

For the IP-based packets and Bluetooth approach, combining the `conn.log` and `flowmeter.log` was no longer a problem, as IP-based packets and Bluetooth only worked with files in the capture format. However, data conversion to CSV format, pre-processing, data cleaning, and traffic labelling were necessary steps to prepare and create both datasets.

Table 4 presents an overview of various information regarding both the IP-based and Bluetooth datasets, including capture duration, number of packets generated, and the number of records within both the packets and flows approaches.

IV. EVALUATION OF ML-BASED INTRUSION DETECTION

We implemented several ML methods to perform binary classification (i.e., identify malicious traffic) and multiclass classification (i.e., recognise the type of traffic, including the type of attack) over the IP-based subsets consisting of packet-based and flow-based data. This section describes the used evaluation metrics and methods, including chosen hyperparameters, and presents the achieved results, followed by a discussion.

A. EVALUATED METHODS AND HYPERPARAMETERS

The models assessed include DNN, Decision Tree (DT), Logistic Regression (LR), Naive Bayes (NB), Random Forest (RF), and Support Vector Machine (SVM). A concise explanation of the techniques used is provided below.

1) DECISION TREE

The DT algorithm is a non-parametric supervised learning method for classification and regression tasks. They partition the feature space into regions, making decisions based on simple rules inferred from the data. Decision trees are interpretable and easy to visualise, having an easily understandable decision-making process.

2) LOGISTIC REGRESSION

Despite its name, LogR is a linear model used for binary classification tasks. It uses a logistic function to predict the probability that a given input belongs to a particular class. LogR is simple and efficient.

3) NAIVE BAYES

NB is a probabilistic classification algorithm founded on Bayes' theorem, making the "naive" feature independence assumption. Despite its simplicity, NB frequently delivers high performance, especially in tasks such as text classification and handling high-dimensional datasets.

4) RANDOM FOREST

RF is a widely employed ensemble learning technique that leverages multiple decision trees to enhance predictive performance. During training, it constructs numerous decision trees and outputs the mode for classification or mean prediction for regression based on the collective decisions of individual trees.

5) SUPPORT VECTOR MACHINE

SVM is another ensemble algorithm used for classification and regression tasks. It works by finding the optimal hyperplane that separates data points belonging to different classes in a high-dimensional space. SVM aims to maximise the margin between the classes, making it robust to outliers and capable of handling complex decision boundaries.

6) DEEP NEURAL NETWORKS

DNNs are a class of artificial neural networks (ANN) with multiple layers between the input and output layers, enabling them to learn complex representations from data through

deep learning. With hidden layers, they automatically capture hierarchical representations from data, thereby capturing increasingly abstract features as information propagates through the network.

The DNN model architecture used for binary classification consists of several densely connected layers with ReLU activation functions, aiming to capture nonlinear relationships within the data. The input layer, comprising 256 neurons, is followed by batch normalisation and a dropout layer to prevent overfitting by randomly deactivating neurons during training. Subsequent hidden layers, consisting of 128, 64, and 32 neurons respectively, also integrate batch normalisation and dropout regularisation techniques. The final layer utilises a sigmoid activation function to produce a binary classification output. The model is optimised using the Adam optimiser and trained to minimise binary cross-entropy loss, while performance is evaluated using accuracy metrics.

On the other hand, the DNN model architecture used for multiclass classification comprises an input layer of 512 neurons, followed by subsequent layers featuring 256, 128, 64, and 32 neurons, each incorporating ReLU activation functions to enable nonlinear transformations of the input data. Batch normalisation and dropout regularisation are integrated after each layer. The final layer, comprised of 9 neurons, employs a Softmax activation function to produce probability distributions across multiple classes. The model is optimised using Stochastic Gradient Descent (SGD) and evaluated based on Sparse Categorical Cross-Entropy loss and accuracy metrics.

Traditional ML algorithms (DT, LR, NB, RF, and SVM) models were imported and initialised with a predefined random state to ensure reproducibility.

Table 5 presents the hyperparameters employed for the various ML models utilised in this study. The DNN hyperparameters encompass both architectural and training parameters, providing a detailed understanding of the DNN developed and applied in this research. Notably, the parameter tuning techniques employed are consistent with those used during dataset construction, as described in Section III-D.

TABLE 5. Hyperparameters of the ML models. If not explicitly specified, default parameters are assumed.

Model	Hyperparameters
DT	Criterion: Gini Max. Leaf Nodes: 5 Random State: 42
LogR	Optimisation Algorithm: lbfgs Random State: 42 Max. Number of Iterations: 5,000
NB	Priors: None Var. Smoothing: 1e-9
RF	Criterion: Gini No. Estimators: 5 Random State: 42
SVM	No. Estimators: 5 Random State: 42
DNN	Architecture: - Activation: ReLU - First Layer Node: Equal to No. of Features - Hidden Layers Nodes (Binary): 256, 128, 64, 32 - Hidden Layers Nodes (Multiclass): 521, 256, 128, 64, 32 - Batch Norm & Dropout (50%) - Final Layer Node: Equal to No. of Classes Training Parameters: - Loss Func. (Multiclass): Sparse Categorical Cross-Entropy - Loss Func. (Binary): Sparse Binary Cross-Entropy - Optimizer: Adam - Callbacks: ReduceLROnPlateau(monitor=loss, factor=0.5) - Epochs: 100 - Batch Size: 512

B. DATASET PRE-PROCESSING

Six the pre-processing methods were employed, namely outlier removal to identify and remove outliers that might skew the data distribution and impact future model performance; data mapping and cleaning to ensure consistency and remove any inconsistencies or errors (refer to Section III-D), one-hot encoding to convert categorical variables into numerical representations facilitating model training, z-score normalisation to normalise numerical features bringing them to a common scale and preventing bias towards specific features, and unnecessary column removal to discard columns that do not significantly contribute to the model's predictive power, reducing dimensionality.

Each dataset was split into training, validation, and testing sets. After the training phase, model predictions were made using the testing set. The training set comprises 75% of the dataset, while the testing set accounts for 25% of the entire dataset, with the validation set representing 25% of the training set.

C. EVALUATION METRICS

We considered several metrics to evaluate the performance of classification models.

1) CONFUSION MATRIX

A confusion matrix is a table that describes the performance of a classification algorithm/model, often referred to as a ''classifier'', when applied to a set of test data for which the actual values are known [78]. The matrix allows for comparing the actual target values with those predicted by the ML model, providing a detailed breakdown of the model's performance [79]. To describe a confusion matrix, the following terms are used: *True Positives* (TP) represent the number of correctly predicted positive instances, while *False Negatives* (FN) indicate the count of actual positive instances that were incorrectly predicted as negative. On the other hand, *False Positives* (FP) signifies the number of actual negative instances that were incorrectly predicted as positive, and *True Negatives* (TN) represents the correctly predicted negative instances by the model [78], [79]. Using the confusion matrix, various performance metrics are computed. These metrics are essential for determining the best model or algorithm's performance within a specific ML application [80].

2) ACCURACY

Accuracy is the ratio between the correctly classified samples and the total number of samples in the evaluation dataset [42], [78], as defined in Equation 1.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

3) PRECISION

Precision is the proportion of retrieved samples that are relevant and is calculated as the ratio between correctly classified samples and all samples assigned to that class [78], as defined in Equation 2.

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

4) RECALL

Also known as sensitivity, recall is the rate of positive samples correctly classified and is calculated as the ratio between correctly classified positive samples and all samples assigned to the positive class [80], as defined in Equation 3.

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

5) F1-SCORE

Defined in Equation 4, the F1-score expresses the harmonic mean of precision and recall. It offers a single metric ranging from 0 (worst) to 1 (best), summarising the performance.

$$F_1\text{-}Score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \tag{4}$$

This metric is not symmetric between classes, meaning it depends on which class is defined as positive and negative [79]. In multiclass classification, evaluating model performance must involve metrics beyond binary classification, which range from 0 to 1 [20].

6) MULTICLASS EVALUATION METRICS

In multiclass classification, one should first compute the metric for each unique class as presented before. Then, one must combine such computed performance values into a single value. The most common ways to do that are using the macro-average and weighted average.

Macro-average treats all classes equally, providing a uniform assessment across the entire classification task. This technique calculates metrics for each class independently and then averages them.

On the other hand, weighted averaging considers the importance of individual values within a dataset by considering the number of instances in each class during the average computation. It is defined by the Equation 5

$$Weighted\ Averaging = \frac{\sum_{i=1}^{n} w_i \cdot x_i}{\sum_{i=1}^{n} w_i} \tag{5}$$

where x_i represents the individual values, w_i represents the weights corresponding to each value x_i, and n is the total number of values.

The weighted averaging assigns distinct importance to individual metric values and may be used when dealing with a dataset where specific data points hold more significance or relevance than others. In this work, we employed weighted averaging when computing the metrics for the multiclass classification.

D. EVALUATION RESULTS

In the following, we present the classification performance of the evaluated ML algorithms.

1) IP-BASED PACKETS DATASET

Table 6 summarises the experimental results for binary and multiclass classification when using the IP-based packets data. The obtained metrics indicate high performance for binary classification, consistently reaching values over 90%. DT, RF, and the DNN achieved recall and precision values of over 0.99. However, in multiclass classification, there is a slight decrease in the overall performance. DT, for instance, achieved a recall of 0.93 and a precision near to 0.97. The worst performance was achieved by NB, whose F1-score was close to 0.83.

TABLE 6. Experimental results of ML applications on the IP-based packets dataset using both binary and multiclass classification.

		Accuracy	Recall	Precision	F1-Score
Binary Class.	DT	99.15	99.15	99.15	99.15
	LogR	92.03	92.03	93.12	92.07
	NB	92.45	92.45	93.44	92.94
	RF	**99.89**	**99.89**	**99.89**	**99.89**
	SVM	92.03	92.03	93.13	92.08
	DNN	99.28	99.28	99.28	99.28
Multiclass Class.	DT	93.89	93.89	97.07	95.19
	LogR	91.50	91.50	93.40	92.44
	NB	79.65	79.65	93.40	83.03
	RF	**99.47**	**99.47**	**99.49**	**99.48**
	SVM	91.50	91.50	92.26	91.88
	DNN	99.07	99.07	99.09	99.08

Figure 3 illustrates the confusion matrices for binary classification for RF (on the left) and LogR (on the right). RF performs admirably, misclassifying 140 packets as normal traffic when they are malicious. However, the LogR matrix reveals a concerning trend, with the algorithm misclassifying 14,931 packets as normal traffic when they are malicious.

Figure 4 illustrates the confusion matrix obtained when applying the NB algorithm for multiclass classification. The figure uses a colouration strength related to the total percentage of each row. While most attacks were accurately predicted, a considerable amount of packets (33,753) representing the normal class were misclassified as resulting from scanning attacks.

2) IP-BASED FLOWS DATASET

Table 7 presents experimental results for both binary and multiclass classification when using the IP-based flows dataset.

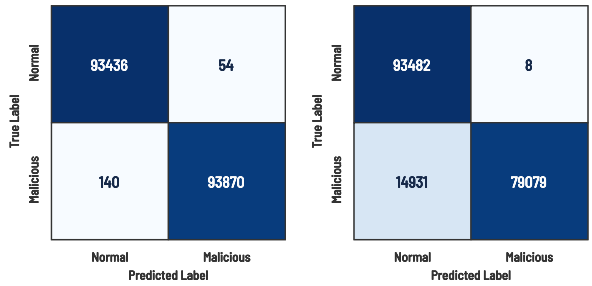


FIGURE 3. Confusion matrices for IP-based packet binary classification using RF and LogR. The left matrix represents the results achieved with RF. The right matrix presents the results achieved with LogR.

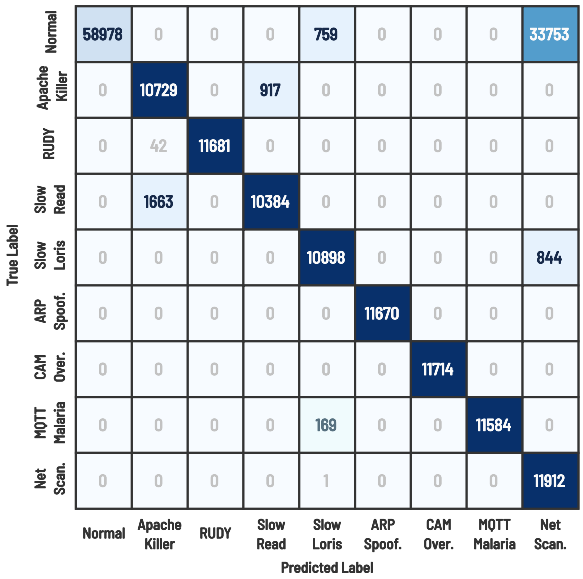


FIGURE 4. Confusion matrix for IP-based packet multiclass classification using NB. Colour intensity indicates the percentage strength of values within each row, emphasising notable distinctions.

TABLE 7. Experimental results of ML applications on the IP-based flows dataset using both binary and multiclass classification.

		Accuracy	Recall	Precision	F1-Score
Binary Class.	DT	**99.90**	**99.90**	**99.90**	**99.90**
	LogR	99.41	99.41	99.41	99.41
	NB	87.90	87.90	90.09	88.98
	RF	99.89	99.89	99.89	99.89
	SVM	99.39	99.39	99.39	99.39
	DNN	99.88	99.88	99.88	99.88
Multiclass Class.	DT	99.45	99.45	99.45	99.45
	LogR	96.95	96.95	97.13	97.04
	NB	92.50	92.50	92.97	92.61
	RF	**99.52**	**99.52**	**99.52**	**99.52**
	SVM	96.93	96.93	96.99	96.96
	DNN	97.63	97.63	97.80	97.71

In binary classification, the performance was overall high. Similar to achieved when using the IP-based packets approach, the NB algorithm performs worse than the other algorithms. Figure 5 illustrates the confusion matrices from the binary classification of the algorithms RF and NB.

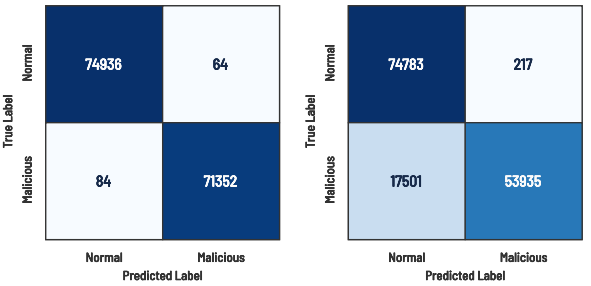


FIGURE 5. Confusion matrices for IP-based flows multiclass binary classification using RF and NB. The left matrix represents the results achieved with RF. The right matrix presents the results achieved with NB.

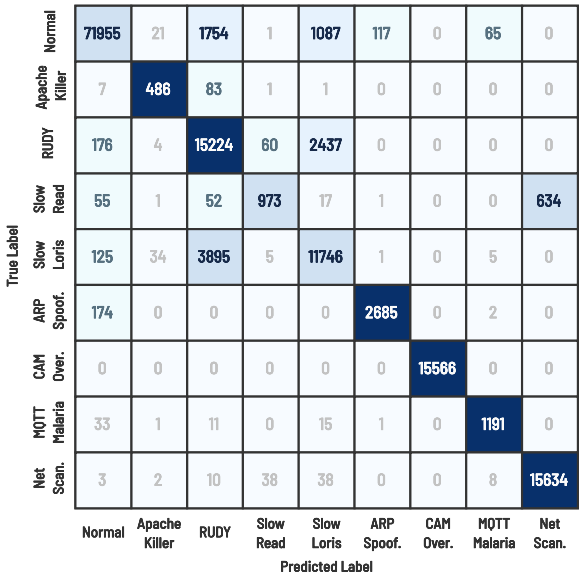


FIGURE 6. Confusion matrix for IP-based flows multiclass classification using NB. Colour intensity indicates the percentage strength of values within each row, emphasising notable distinctions.

One can observe the stark difference in the number of FN between the two approaches. NB exhibits the highest number of misclassified normal traffic packets as malicious, with 17,501 packets misclassified. This contrasts to RF, which misclassifies only 84 packets.

As in the case of the packet-based dataset, the algorithms performed worse for multiclass classification than for binary classification. Figure 6 presents the confusion matrix resulting from the application of the NB algorithm, which achieved a considerable number of misclassified flows for the DoS variants, especially the Slow Loris, Slow Read, and the Rudy attacks. Indeed, the algorithm misclassifies 634 flows of Slow Read as Net Scanning attacks and correctly identifies the attack in only 973 flows, resulting in a per cent error of 65.16%.

E. DISCUSSION

Table 8 summaries various classification metrics within the algorithms and models implemented. It is clear that for IP-based packets, the algorithm that presents better performance is RF, with metric values consistently at 99.89%.

For an IP-based flows approach, DT presented better results with metric values consistently at 99.90%.

TABLE 8. Experimental results of ML applications within both IP-based packets and IP-based flows datasets with binary classification, aggregated by the employed classification metrics: accuracy, recall, precision, and F1-score.

		DT	LogR	NB	RF	SVM	DNN
Packets	Accuracy	99.15	92.03	92.45	**99.89**	92.03	99.28
	Recall	99.15	92.03	92.45	**99.89**	92.03	99.28
	Precision	99.15	92.12	93.44	**99.89**	92.13	99.28
	F1-Score	99.15	92.07	92.94	**99.89**	92.08	99.28
Flows	Accuracy	**99.90**	99.41	87.90	99.89	99.39	99.88
	Recall	**99.90**	99.41	87.90	99.89	99.39	99.88
	Precision	**99.90**	99.41	90.09	99.89	99.39	99.88
	F1-Score	**99.90**	99.41	88.98	99.89	99.39	99.88

LogR got a recall of 92.03% and a precision of 92.12% when using the IP-based packets and achieved 99.41% in all metrics when using the IP-based flows. SVM achieved performance quite similar to the one of LogR. NB presented the worst results across all metrics. In the IP-based packets dataset, NB got a recall of 92.45% and a precision of 93.44%, and it got a recall and precision of 87.69% and 90.09%, respectively, for the IP-based flows.

In multiclass classification, the results are slightly different from those of binary classification. Table 9 summarises the classification metrics values within the algorithms and models implemented for multiclass classification.

RF presents the best results for both the IP-based packets and IP-based flow datasets. It achieved 99.52% in all metrics for the IP-based flows dataset. When using the IP-based packets dataset, RF exhibits a precision of 99.49%, accuracy and recall of 99.47%, and F1-score achieved 99.48%.

TABLE 9. Experimental results of ML applications within both IP-based packets and IP-based flows datasets with multiclass classification, aggregated by the employed classification metrics: accuracy, recall, precision, and F1-score.

		DT	LogR	NB	RF	SVM	DNN
Packets	Accuracy	93.89	91.50	79.65	**99.47**	91.50	99.07
	Recall	93.89	91.50	79.65	**99.47**	91.50	99.07
	Precision	97.07	93.40	93.44	**99.49**	92.26	99.09
	F1-Score	95.19	92.44	83.03	**99.48**	91.88	99.08
Flows	Accuracy	99.45	96.95	92.50	**99.52**	96.93	97.63
	Recall	99.45	96.95	92.50	**99.52**	96.93	97.63
	Precision	99.45	97.13	92.97	**99.52**	96.99	97.80
	F1-Score	99.45	97.04	92.61	**99.52**	96.96	97.71

LogR and SVM present similar results. For IP-based packet data, LogR and SVM achieved the same recall value (91.50%), and their precision was 93.40% and 92.26% respectively. When using the flows dataset, LogR achieved a recall of 96.95% and 97.13% for precision, while the recall for SVM was 96.93% and its precision was 96.99%. NB presents the worst results among all of the algorithms deployed. It exhibits a recall of 79.65% and a precision of 93.44% within the IP-based packets dataset and a recall of 92.50% and a precision of 92.97% for IP-based flows. The DNN achieved 99.07% and 99.09% for recall and precision, respectively, on the IP-based packets dataset, and 97.63% and 97.80% for recall and precision when using flow data.

Two algorithms particularly excel in comparison to the others: DT and RF. Both algorithms consistently achieve recall values near 99%. Recall may be considered a key performance indicator in the context of the ML-based IDS. This emphasis on recall is due to the need to minimise instances where the model incorrectly predicts malicious traffic as normal. These instances represent potential attacks mistakenly labelled as harmless. Therefore, having higher recall is crucial as it significantly contributes to accurately identifying and flagging malicious activities, even if it comes at the expense of slightly lower precision. Additionally, both DT and RF demonstrate the lowest training time compared with other algorithms and models, which is advantageous. While DT and RF perform exceptionally well, NB performs poorly compared to the other algorithms. It achieves a recall of just 79.65%. Predicting a significant amount of malicious traffic as harmless could severely compromise the network's security. LogR and SVM consistently demonstrate reasonable performance. On the other hand, the DNN presents excellent results, achieving more than 97% in recall. However, it has higher complexity and training time than the other evaluated algorithms.

The IoMT-TrafficData also enables a comparison between the use of packet-based data and flow-based data for network traffic classification. Table 10 presents the percentage variance between flows and packets regarding recall, precision, and F1-score.

TABLE 10. Performance increment when using flows instead of packets. The outcomes denote the percent variance between flows and packets in terms of recall, precision, and F1-score.

	Binary Class.			Multiclass Class.		
	Rec.	Pre.	F1	Rec.	Pre.	F1
DT	00.75	00.75	00.80	05.56	02.38	04.26
LogR	07.38	06.29	07.34	05.45	03.73	04.60
NB	-04.55	-03.35	-03.96	12.85	-00.43	09.58
RF	00.00	00.00	00.00	00.05	00.03	00.04
SVM	07.36	06.26	07.31	05.43	04.73	05.08
DNN	00.60	00.60	00.60	-01.44	-01.29	-01.37
Average	01.92	01.75	02.01	04.56	01.53	03.70

In binary classification, except for NB and RF, all evaluated algorithms got a variance favouring traffic-based flows, with a maximum variance of 7.38% achieved using LogR. On average, recall, precision, and F1-score demonstrated better results for flows, achieving positive variances of 1.92%, 1.75%, and 2.01% respectively. For NB, a variance favouring traffic-based packets was observed, achieving 4.55%, 3.35%, and 3.96% in recall, precision, and F1-score respectively. Remarkably, RF demonstrated identical performance in both approaches.

In multiclass classification, a favourable deviation is noted for traffic-based flows, akin to binary classification. All algorithms, except for DNN, exhibited a positive deviation for traffic-based flows, notably displaying a 12.85% deviation with NB concerning recall. However, with the same NB, the traffic-based flows approach demonstrated a negative deviation of 0.43% for precision, indicating some instability in NB results. Concerning RF, a slight positive deviation in favour of flows, ranging from 0.03% to 0.05%, was observed. Regarding DNN, all evaluation metrics exhibited a positive deviation for the traffic-based packets approach, yielding results of 1.44%, 1.29%, and 1.37% for recall, precision, and F1-score, respectively. On average, all evaluation metrics indicate a positive deviation concerning traffic-based flows, peaking at 4.56% deviation for both recall and F1-score.

Indeed, the traffic-based flows approach yielded the most favourable outcomes, achieving up to 2% better results for binary classification and an increment in performance of almost 5% for multiclass classification. Moreover, employing flows in deploying models within an IDS network may reduce the data transfer volume between acquisition points and the IDS.

V. CONCLUSION

In the contemporary world, where intelligent information systems are implemented across various domains, medical systems are no exception. Interconnected systems exchange valuable data about the medical infrastructure but also patient data. Therefore, if adversaries carry out attacks and gain access to this data, they may compromise integrity or disrupting the medical infrastructure, thus threatening availability. Within this problem, various solutions arise. One very promising solution is the integration of ML with IDS. For this purpose, traffic analysis must be conducted, and test scenarios must be constructed to capture traffic generated by both malicious and benign sources.

In this paper, we present a new dataset for evaluating intrusion detection systems on the IoMT. To build the dataset, we developed a representative IoMT scenario, conducted various attacks on the developed scenario, collected traffic generated from these attacks and from normal behaviour, and organised collected data into packet-based and flow-based subsets. We also applied various ML models and algorithms to identify malicious traffic using data on gathered packets and network flows. DT and RF achieved high performance, with values over 95% for recall and precision. Additionally, we showed that using traffic-based flows is preferable to traffic-based packets for most algorithms, as a flows-based approach achieved up to 2% better results for binary classification and almost 5% better results for multiclass classification. Obtained classification results may serve as a baseline for future benchmarking and evaluation of new intrusion detection models.

In future studies, we plan to investigate the applicability of novel models and distributed learning architectures in the IoMT environment, using the *IoMT-TrafficData* dataset for training and testing such models and compare the achieved performance with the ones in this paper. We also aim to expand the dataset, incorporating additional attack types and traffic data for more protocols representative of the IoMT environment, specifically those from hospital equipment.

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